

Predicting California Housing Prices

Final Project Report

Dmitrii Shurkhai

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1. Project Overview

The goal of this project is to predict **housing prices in California districts** using machine learning.

Housing prices depend on several factors such as location, income level, and housing characteristics. By analyzing these factors, we can build a model that estimates house values.

Business Question

The main question for this project is:

Which factors affect housing prices the most, and can we predict house values using this information?

Hypothesis

We expect that the following factors will influence housing prices:

- Median income
- Geographic location
- Housing characteristics
- Distance from the ocean

To test this, a **Random Forest regression model** was used.

2. Description of the Data Set

The dataset used in this project is the **California Housing dataset**, based on data from the **1990 U.S. Census**.

Each row represents a **district in California**.

The dataset includes information about housing, population, income, and geographic location.

Variables

Variable	Description
longitude	Geographic longitude
latitude	Geographic latitude
housing_median_age	Median age of houses
total_rooms	Total number of rooms
total_bedrooms	Total number of bedrooms
population	Number of residents
households	Number of households
median_income	Median income of residents
ocean_proximity	Location relative to the ocean
median_house_value	Median house value

The **target variable** we want to predict is:

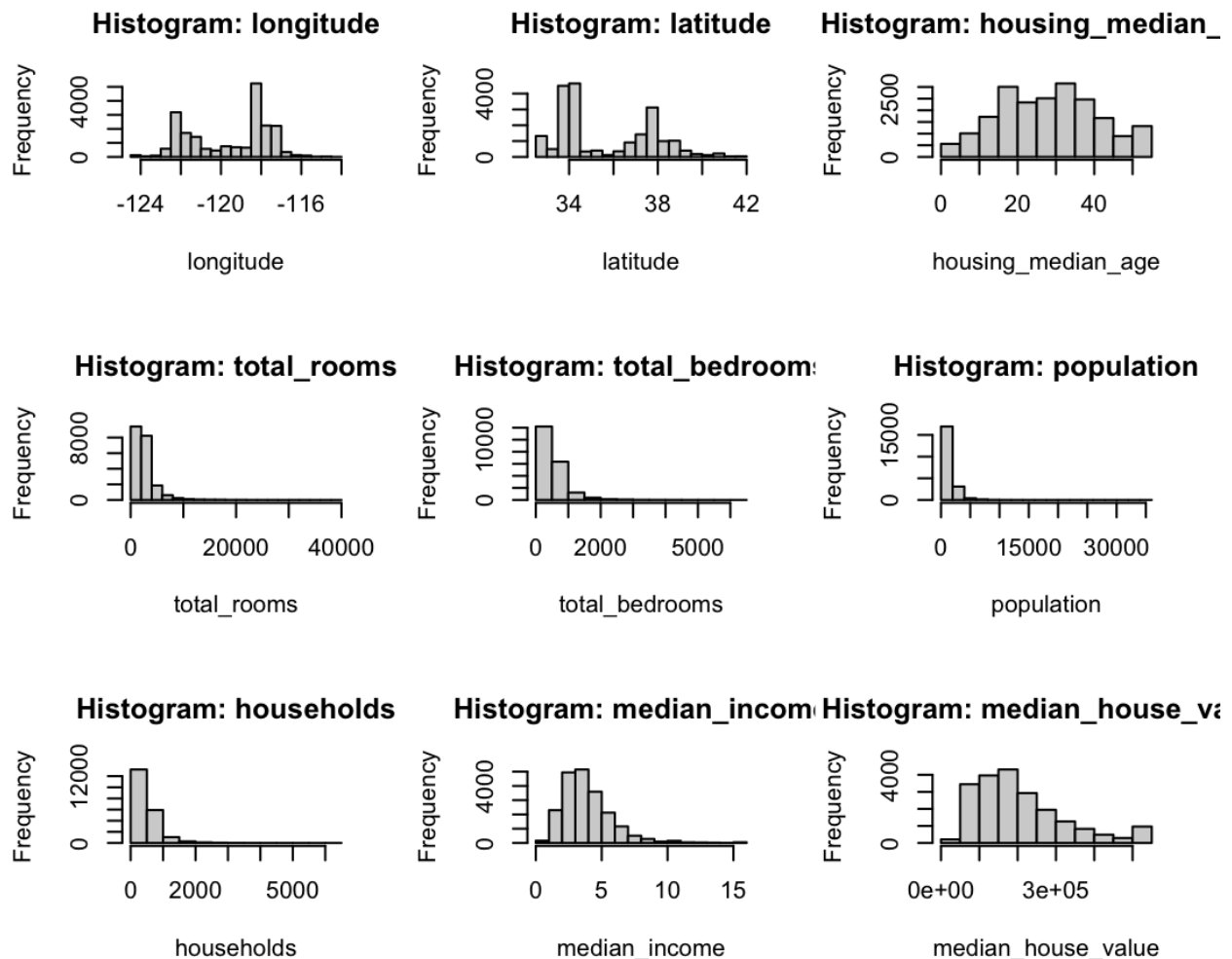
median_house_value

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify patterns.

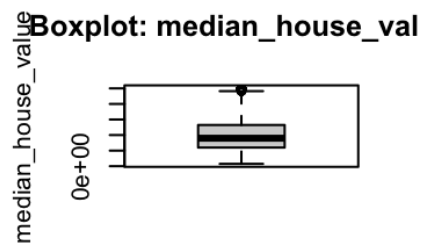
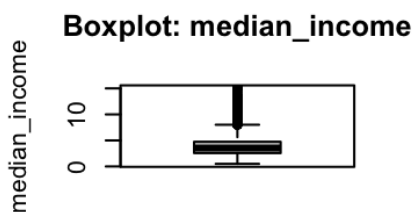
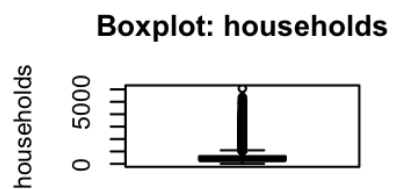
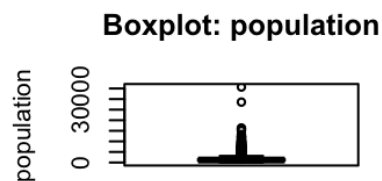
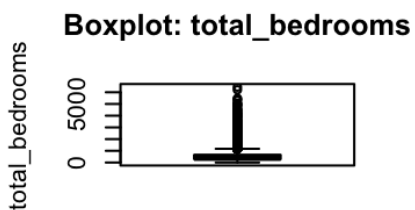
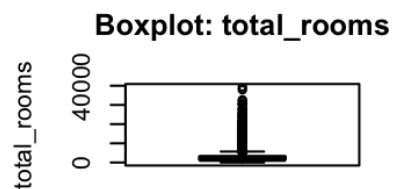
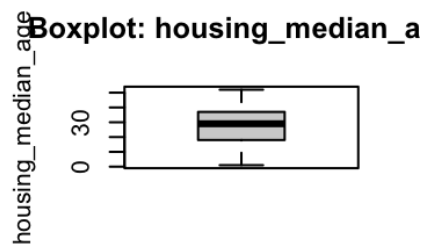
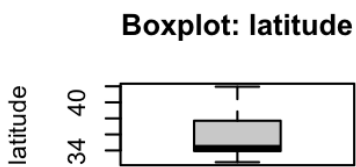
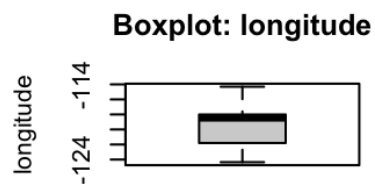
First, summary statistics were generated for all variables.

Next, **histograms** were created to examine the distribution of numeric variables.

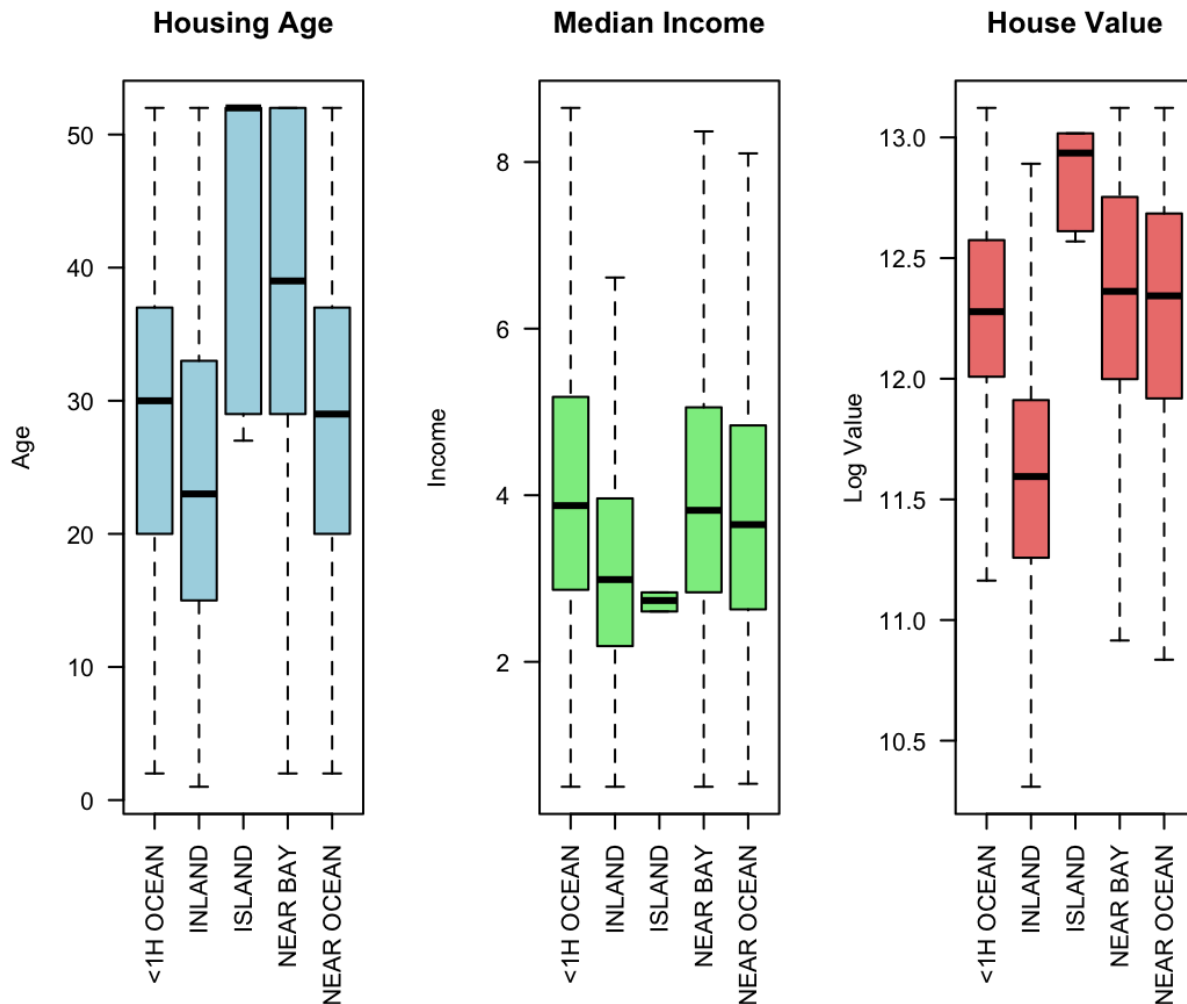


The histograms show that some variables such as population and number of rooms have a **skewed distribution**, meaning a few districts have very large values.

Boxplots were also created to detect outliers.



Finally, boxplots were used to compare variables across **ocean proximity categories**.



The results suggest that districts closer to the ocean often have **higher housing prices**.

4. Data Preparation

Before building the model, several preprocessing steps were applied.

Missing Values

The variable **total_bedrooms** contained missing values.

These values were replaced using the **median value** of the variable.

Encoding Categorical Data

The variable **ocean_proximity** was converted into binary variables (one-hot encoding).

Feature Engineering

Two new variables were created:

- **mean_bedrooms** = total bedrooms divided by households
- **mean_rooms** = total rooms divided by households

These features represent the **average number of rooms per household**.

The original total room variables were then removed.

Feature Scaling

Numeric variables were standardized so that they have similar ranges.

Train/Test Split

The dataset was split into:

- **70% training data**
- **30% testing data**

The training data was used to build the model, and the test data was used to evaluate its performance.

5. Statistical Model

A **Random Forest regression model** was used.

Random Forest works by building many decision trees and combining their predictions.

This approach improves prediction accuracy and reduces overfitting.

Model Inputs

The model used features such as:

- location (longitude and latitude)
- population and households
- median income
- ocean proximity
- engineered variables (mean_rooms and mean_bedrooms)

Response Variable

The model predicts:

median_house_value

The model was trained using **500 trees**.

6. Model Performance

Model performance was evaluated using **Root Mean Squared Error (RMSE)**.

RMSE measures how far predictions are from the true values.

Training RMSE

49512

Test RMSE

48304.96

Considering that housing prices in the dataset range roughly from **\$15,000 to \$500,000**, an average prediction error of about **\$48-49k** is reasonable for a real-world housing prediction model.

The results show that the Random Forest model is able to **capture important patterns in the data and provide reliable predictions of housing prices.**

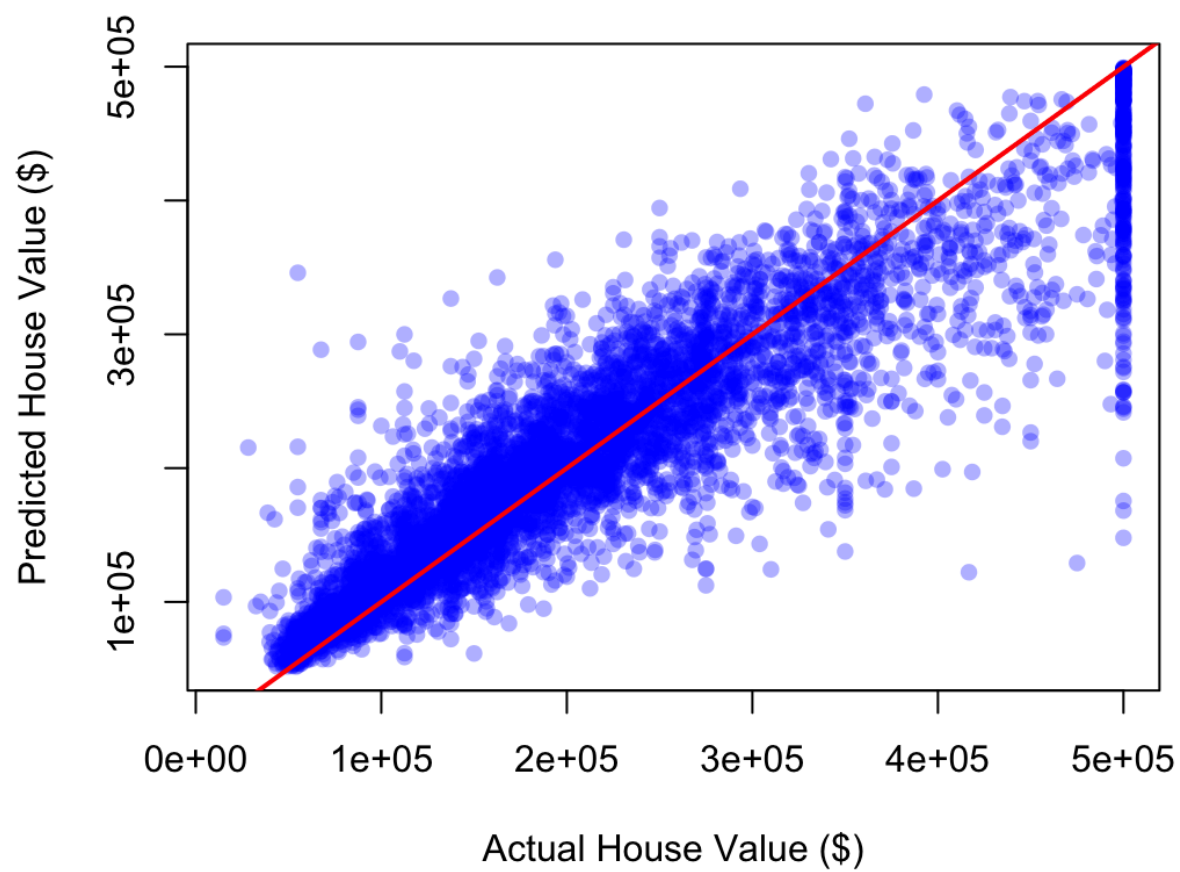
Additional Model Evaluation

As an additional step beyond the project requirements, a scatter plot of **predicted versus actual housing prices** was created to further evaluate the performance of the Random Forest model. This visualization compares the model's predictions on the test dataset with the true housing prices. The red diagonal line in the plot represents perfect predictions, where the predicted value is equal to the actual value.

Most data points follow the general diagonal trend, indicating that the model is able to capture the relationship between the predictors and housing prices reasonably well. The spread of points around the line reflects the prediction error measured by the RMSE metric. A vertical concentration of points near \$500,000 can also be observed, which occurs because the dataset caps house values at this level.

Although this visualization was not required for the project, it provides an additional way to confirm that the Random Forest model produces reasonable predictions and captures the overall pattern in the data

Predicted vs. Actual Housing Prices



7. Business Answer

The results show that **income and location are the most important factors affecting housing prices.**

Key findings:

- Districts with **higher median income tend to have higher house prices**
- **Location near the ocean increases housing values**
- Housing density and number of rooms also influence prices

Conclusion

The Random Forest model successfully predicts housing prices using demographic and housing data.

These insights can help:

- real estate investors
- housing analysts
- policy makers

better understand housing price patterns in California.